The Short, the Long, and the Consequences of VAR Analysis for Structural DSGE Estimation *

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Abstract

De Graeve and Westermark (2025) advocate the use of long lags VARs for structural analysis, claiming that the truncation bias is a severe problem for the IRFs estimation. VARs can be used to create additional moments to estimate DSGE models minimizing the distance between their IRFs and therfore estimating the underlying structural parameters. In this study we ask what are the consequencies of enriching the lag structure of a VAR used for DSGE estimation. I replicate the results of Altig et al. (2011) and I find that the both IRFs and Structural parameters are sensible to the chosen lag lenght. Matching the DSGE with a VAR(12) instead of the baseline VAR(4), I find that the Calvo Price parameters that differencieate the observationally equivalent Altig et al. (2011) model between firm specific capital vs homogeneous capital are no longer statistically different from each other. The implied duration of the Calvo Price parameter is 2.5 times higher than the baseline, re opening the micro-macro discrepancy in price stickiness.

Keywords: Truncation Bias, Long Lag VARs, DSGE Estimation, Minimum Distance, IRFs
JEL Codes: C32, C52, E32

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1 Introduction

Introduzione è la parte più importante e difficile, magari la abbozzo e poi la riscrivo alla fine.

1.1 Research Question

I analyse the consequences of enriching the lag structure of a VAR used to estiamte a DSGE with IRFs minimum distance estimation. I replciate Altig et al. (2011) and find that the both IRFs and Structural parameters are sensible to the chosen lag lenght.

1.2 Main Contributions

The Calvo Price parameters that differencieate the observationally equivalent Altig et al. (2011) model between firm specific capital vs homogeneous capital are no longer statistically different from each other. This go against their main conclusion that the firm specific capital model is the one that better fit the data.

(Maybe the Capital Adjustment Cost variable that since Altig et al. (2011) researchers have started to use, is not fundamental to explain inflation intertia.)

1.3 Related Literature

Long Lag VARs De Graeve and Westermark (2025); Baumeister (2025); Antolín-Díaz and Surico (2025); Antolín-Díaz et al. (2026)
IRFs (VARs and LPs) Ludwig (2026); Li et al. (2024); Òscar Jordà (2005)
State Space rappresentation of DSGE Ravenna (2007); Fernández-Villaverde et al. (2005); Forni et al. (2016); Theodoridis (2011); Giacomini (2013)

1.4 Outline

The rest of the work is organised as follows. In Section 2 I present the Truncation Bias problem, the theory that maps DSGE into VAR(inf) that suggests to increase the lag lenght. In section 3 I summaryse breafly the Altig et al. (2011) DSGE. In section 4 I present the empirical results and in section 5 I conclude.

2 Theoretical Framework

The Truncation Bias problem, expalind easy as De Graeve and Westermark (2025).

Simple theoretical derivation describing that even if the Altig et al. (2011) is stochastich singular (and defend the fact that the most of the SVAR analysis is done with stochastig singular models (more observables than shocks), Forni et al. (2016)) and explain that the solution of the DSGE in State Space can be representate as a VARMA Ravenna (2007), hence the use of VAR with long lags is justified even if the ABC and D Fernández-Villaverde et al. (2005) representation of the VAR(∞) does not exist.

3 Theoretical Model

Then a breif descriptio of the Altig et al. (2011) DSGE using the replication package from (ALEL NBER 2005 WORKING PAPER)

4 Empirical Results

4.1 Implied Duration from Calvo Price Table: lags 4 8 12

Table 1: Implied Average Time between Re-Optimization

Model Specification	**VAR** $p = 4$	**VAR** $p = 8$	**VAR** $p = 12$
Homogeneous Capital Model			
Calvo parameter ($\xi_{p,hom}$)	0.82	0.82	0.90
Implied duration (quarters)	**5.6**	**5.6**	**9.6**
Firm-Specific Capital Model			
Calvo parameter ($\xi_{p,fs}$)	0.34	0.64	0.89
Implied duration (quarters)	**1.5**	**2.8**	**9.0**

- **Baseline VAR** ($p = 4$): Homogeneous capital implies price changes every **5.6 quarters**.
- **Firm-specific capital**: Drastically reduces this to **1.5 quarters**, matching microeconomic evidence.

Note: Implied duration is calculated as $D = 1/(1 - \xi_p)$.

5 Conclusion

References

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A Replication Files and Code

Detail where the reader can find your code (e.g., a GitHub link).

B Additional Tables

C Additional Figures

C.1 ACEL 2005 VAR Variables

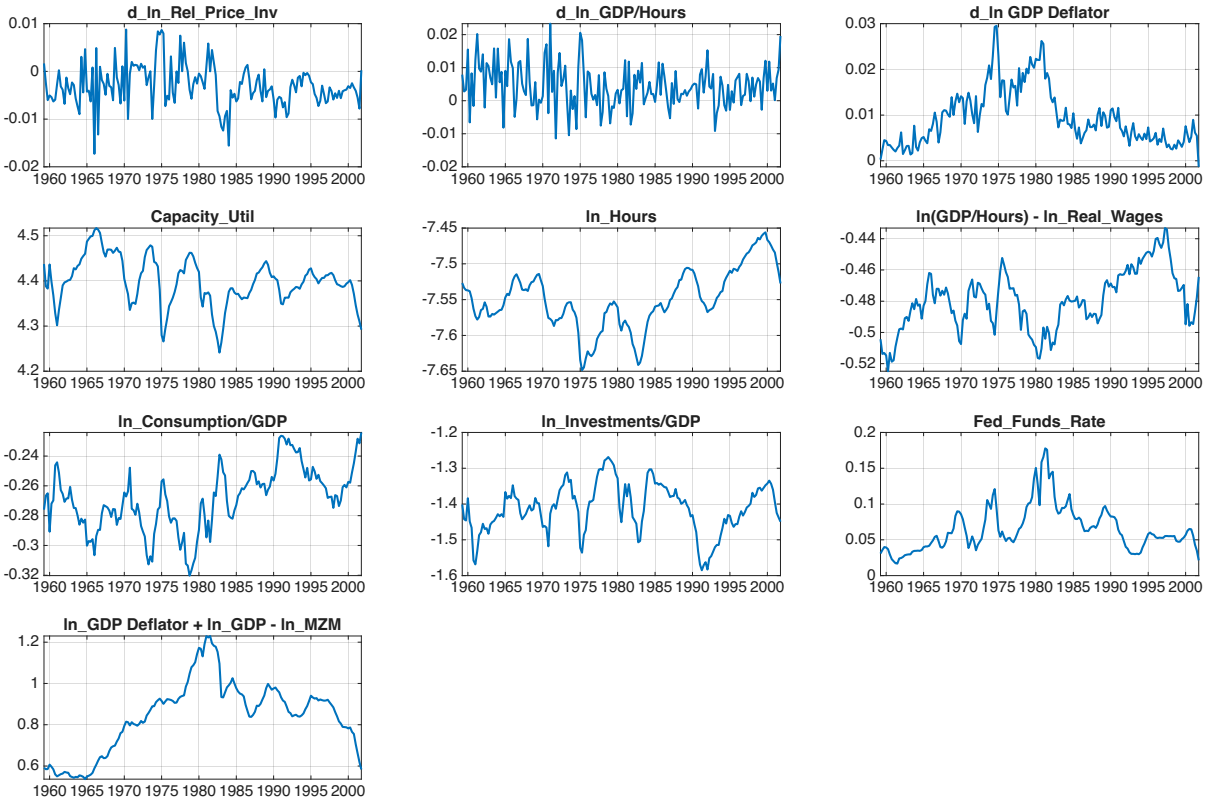


Figure 1: Raw Historical Series